

Deployable Symmetry-Coset Conditioning for Orientation Flows in Molecular-Crystal Structure Prediction

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Rigid-body flow matching is an efficient route to molecular-crystal structure prediction, reducing a crystal to a lattice, fractional centroids, and per-molecule orientations on $\text{SO}(3)$. Recent rigid-body molecular-crystal flows leave space-group symmetry unconditioned, and symmetry-conditioned generation has so far been demonstrated only for inorganic atomic sites, never for rigid-body molecular orientation. We show that the per-molecule orientation target decomposes as $R_m = \text{rot}(g_m) R_{\text{asym}}$ into a space-group-determined relative rotation, which a flow can learn, and a gauge-free asymmetric-unit pose, which it cannot regress from packing. We turn this into a deployable method: a *leak-free* coset label—the generating space-group operation, recoverable from a template at sampling time—conditions the orientation flow. On 1,127 crystals from the Cambridge Structural Database, deployable coset conditioning lifts the held-out non-reference orientation loss by 41.1% (versus 27.5% for a paired no-coset control), recovering two-thirds of the way to a leaky-codebook oracle ($\sim 48\%$); an $\text{SO}(3)$ -averaged training objective closes the remaining gap (47.7%). A packing-only predictor recovers the coset at 39.5% accuracy ($4\times$ the majority baseline) but is not yet accurate enough for template-free use, which we identify as the key remaining gap. The method converts the inference-limited part of the orientation field into a near-deterministic, template-available lookup, leaving only the free pose to be sampled.

1 Introduction

Crystal structure prediction (CSP) asks for the stable periodic arrangement of a chemical system. Generative models have become a fast alternative to large ensembles of density-functional relaxations, progressing from crystal diffusion autoencoders[1] to joint equivariant diffusion[2] and Riemannian flow matching[3–5], which learns a straight conditional path solvable in tens of steps. *Rigid-body* factorization is especially attractive for molecular crystals: freezing each molecule’s conformer collapses dozens of atomic coordinates to a single pose—a lattice L , a fractional centroid $c_m \in \mathbb{T}^3$, and an orientation $R_m \in \text{SO}(3)$ per molecule—and supports the short sampling trajectories characteristic of flow matching.

This introduces a per-molecule orientation variable on $\text{SO}(3)$, and here the current generation of models leaves value on the table. MolCrystalFlow, a 2026 rigid-body molecular-crystal flow with the same $\text{lattice} \times \mathbb{T}^3 \times \text{SO}(3)$ factorization, jointly learns pose but *does not impose space-group constraints*[6]; MOFFlow likewise flows rigid $\text{SE}(3)$ poses without a symmetry prior, relying on framework connectivity to constrain orientation[7]. Symmetry- and Wyckoff-conditioned generation, meanwhile, is proven only for *inorganic atomic sites*: DiffCSP++[8], WyckoffDiff[9], SymmCD[10], and the 2026 NextCrystal[11] all condition atom generation on a space-group/Wyckoff template, but none touches rigid-body molecular pose. The gap is precise: nobody has conditioned a rigid-body *molecular orientation* flow on the crystal’s symmetry.

We close that gap. The starting point is a decomposition of the orientation target, $R_m = \text{rot}(g_m) R_{\text{asym}}$, into a space-group-determined relative rotation between symmetry copies and the asymmetric unit’s gauge-free pose. Classical molecular-CSP codes never search the relative rotation—they generate symmetry copies by applying space-group operators[12, 13]—and we find a learned flow mirrors that division of labor: the relative part is learnable and the free part is not. The contribution of this paper is to make the learnable part *deployable*. Our contributions are:

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- A **leak-free, template-available coset label**—the generating space-group operation $h = g_m g_0^{-1}$, recovered from centroids alone—that supplies the symmetry-relative rotation at sampling time without seeing the target (Section 2), in contrast to a codebook clustered from the target itself.
- A coset-conditioned orientation flow and sampler that **recover two-thirds of the oracle-codebook benefit** (+41.1% vs a +27.5% paired control on held-out non-reference orientation loss; Section 3).
- An **SO(3)-averaged training objective**[14] that closes the remaining gap to the leaky ceiling (+47.7%).
- A **packing-only coset predictor** for template-free use (39.5% top-1, $4\times$ majority) and an honest characterization of the remaining de-novo gap.

2 Method

Rigid-body flow. SymMC-Flow represents a molecular crystal by a lattice $L \in \mathbb{R}^{3 \times 3}$ and, per molecule m , a fractional centroid $c_m \in \mathbb{T}^3$ and orientation $R_m \in \text{SO}(3)$ acting on a shared body-frame conformer, so $\text{cart}_{m,a} = c_m L + R_m \text{local}_{m,a}$. A conditional flow-matching objective[4] is trained jointly over three manifolds (a Euclidean path for a volume-normalized lattice, a torus geodesic for centroids, an $\text{SO}(3)$ geodesic for orientations), with the velocity field conditioned on the space group. An $E(n)$ -equivariant encoder[15] embeds the conformer and pair-bias attention mixes molecules within the cell (Appendix A).

The orientation decomposition. A molecule m generated from the asymmetric unit by an operation with rotation part $\text{rot}(g_m)$ has absolute pose $R_m = \text{rot}(g_m) R_{\text{asym}}$, where R_{asym} is the free orientation of the asymmetric unit. Because the body-frame gauge is arbitrary, R_{asym} carries no consistent signal tied to composition or packing: trained directly, the absolute orientation head never beats its predict-zero floor ($\mathbb{E}\|u_R\|^2 \approx 5.24$), while lattice and centroid heads converge normally. Re-gauging each crystal so the first copy of a species is a reference ($R=I$) and the others carry only the relative rotation $R'_m = R_m R_0^{-1}$ cancels R_{asym} and exposes the learnable factor; the held-out non-reference loss then drops by 27%. This much is a diagnostic; the rest of the paper makes it a method.

Deployable symmetry-coset conditioning. Within a space group the non-reference relative rotations fall into a small discrete set of *cosets*: R'_m equals the rotation part of the operation $h = g_m g_0^{-1}$ that maps the reference copy to copy m . We assign each molecule this generating operation directly from the crystallography—by matching its fractional centroid against $W_h c_0 + t_h$ (minimum image) over the space group’s operations $\{(W_h, t_h)\}$ obtained from pymatgen—never from the observed rotation. The resulting label is **leak-free** and available at sampling time from a symmetry template, exactly as DiffCSP++/WyckoffDiff-style generators supply Wyckoff templates for atoms[8, 9]. This yields 238 operation-cosets across the corpus (versus 707 for a codebook clustered from the target rotations, whose index is derived in part from the answer and therefore serves only as an upper bound). The coset index enters the network as a per-molecule embedding and is threaded through the sampler, so it conditions both training and generation.

SO(3)-averaged objective. The $\text{SO}(3)$ flow-matching residual has high variance because the prior rotation is drawn uniformly. Following the $\text{SO}(3)$ -averaged flow objective for conformer generation[14], we average the orientation residual over K prior draws per molecule before the gradient step, reducing target variance at fixed data.

Packing-only predictor. For template-free deployment the coset must be inferred. We train a classifier that predicts each molecule’s coset from the conformer, centroid, lattice, and space group—*without* orientation, so it cannot leak the target—and use its prediction in place of the template at sampling time.

3 Experiments

Setup. The corpus is a seeded CSD sample[16]: 1,127 ordered organic molecular crystals, 5,048 rigid blocks, a fixed 964/131 split, with 1,095 crystals containing ≥ 2 copies of a species (the multi-copy subset where a relative target exists). The primary metric is the *held-out non-reference orientation-loss drop* (percentage reduction from

Table 1: Held-out non-reference orientation-loss drop (untrained→trained), mean±s.d. over three seeds. The deployable coset is leak-free and available from a template; the leaky codebook (index clustered from the target) is an upper bound. Deployable conditioning recovers two-thirds of the gap; the SO(3)-averaged objective closes it.

configuration	non-ref. loss drop	deployable?
no-coset control (baseline)	$27.5 \pm 2.7\%$	—
deployable coset ($K=1$)	$41.1 \pm 3.4\%$	yes
+ SO(3)-averaged ($K=4$)	$47.7 \pm 3.9\%$	yes
leaky codebook (oracle upper bound)	$\sim 48\%$	no

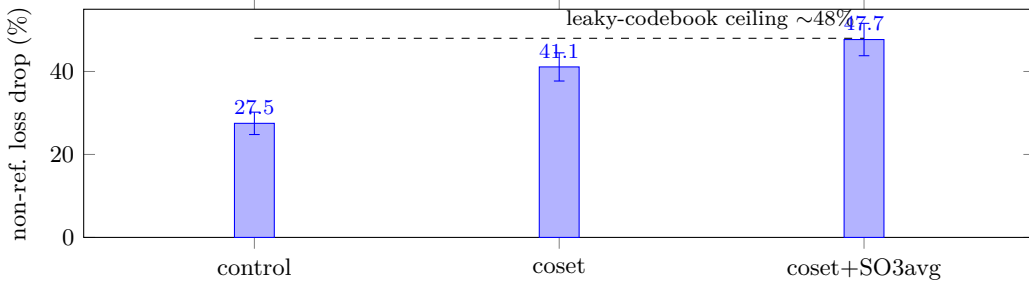


Figure 1: Deployable coset conditioning (leak-free, template-available) recovers two-thirds of the way from the no-coset control to the leaky-codebook oracle ceiling; the SO(3)-averaged objective closes the rest. Three-seed mean; error bars s.d.

untrained to trained on non-reference copies, which cannot be earned by predicting the identity); we also report orientation-isolated StructureMatcher[17] reconstruction (true lattice/centroid/conformer, only SO(3) sampled, best of eight draws, following match-rate cautions[18]). All numbers are means over three seeds.

Deployable coset conditioning recovers most of the oracle benefit. Table 1 and Fig. 1 give the main result. A paired no-coset control reproduces the +27.5% baseline. Supplying the deployable, leak-free coset raises the non-reference drop to +41.1%—recovering about two-thirds of the way from the baseline to the $\sim 48\%$ reached by the leaky codebook that peeks at the target. Adding the SO(3)-averaged objective ($K=4$) reaches +47.7%, statistically level with the leaky ceiling. In other words, the benefit that a target-derived codebook could only *upper-bound* is realized here by a label a sampler actually has.

The conditioned flow reconstructs real packings. A loss drop need not be geometry. Holding the true lattice, centroid, and conformer fixed and sampling only the coset-conditioned SO(3) field, the flow reconstructs held-out multi-copy packings exactly under StructureMatcher in 9.9% (single draw) to 27.5% (best of eight) of cases, rising to 37.4% at looser but oracle-validated tolerances, with a median non-reference orientation error of 41° ; the oracle (true orientation) matches every structure. The symmetry-relative rotations cluster near 180° (the dominant $P2_1/c$, $P\bar{1}$ operations), so best-of- k is the correct read of a multimodal target.

Template-free prediction is the remaining gap. The packing-only predictor recovers the generating coset at 39.5% top-1, $4\times$ the 10.3% majority baseline—packing carries real signal about the symmetry operation. It is not yet enough: substituting the *predicted* coset into orientation-isolated reconstruction collapses to a 64.8° error, worse than the 41° of the template-supplied coset and the no-coset baseline, because a wrong coset points to the wrong target rotation. The deployable gain is thus real whenever a symmetry template is available (the template-based CSP regime), and closing the template-free gap—via a stronger or top- k -calibrated predictor—is the natural next step.

Scope: orientation is not the end-to-end bottleneck. Sampling all three manifolds from the prior (full de-novo generation) yields no exact match at this corpus scale (0%, Wilson 95% CI 0–2.8%). The best-of- k

error budget locates the bottleneck away from orientation: median lattice-parameter error 0.44 and fractional-centroid error 0.35 dominate, while the orientation error is only 13.4°. Our contribution is therefore scoped to the orientation mechanism, which this work advances; joint lattice-centroid generation at scale remains the limiting factor, consistent with the modest ($\sim 1,000$ -crystal) training budget.

4 Discussion

The learnable content of a rigid-body orientation field is the space-group-determined relative rotation, and it can be delivered by a deployable, leak-free coset label that a template-based sampler already possesses; an $SO(3)$ -averaged objective brings this to the level of an oracle codebook that peeks at the answer. This gives a concrete recommendation for rigid-body molecular-crystal generators such as MolCrystalFlow[6], which currently omit symmetry conditioning: condition orientation on the generating coset—turning the inference-limited part of the field into a near-deterministic lookup—and sample, rather than regress, the remaining gauge-free asymmetric-unit pose. The main open problems are template-free coset prediction (here 39.5%, not yet sufficient) and end-to-end lattice-centroid generation, which bounds exact reconstruction independently of orientation. The analysis is scoped to rigid, largely organic molecules on general positions; molecules on special positions carry a site symmetry that further constrains the free pose and should be even more favorable.

Reproducibility. Code, the CSD refcode manifest, and the export pipeline are at <https://github.com/fronkt/symmc-flow>, archived at Zenodo[19]. CSD structures are licence-restricted and not redistributed; a manifest reproduces the corpus from a licensed installation.

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A Method and training details

Intrinsic gauge. Each crystal is parsed into rigid blocks by a covalent-bond graph with periodic unwrapping; species are keyed by a Weisfeiler–Lehman hash and share one reference conformer rotated into a molecule-intrinsic frame (gyration-tensor principal axes, sign-fixed by an element-weighted third moment). Each copy’s pose is the Kabsch rotation to that frame, removing the cross-crystal reference that otherwise floors R_m .

Manifolds. Lattice in a 10-d log-volume/unit-determinant-shape parametrization (prior volume matched to the corpus mean); centroids on torus geodesics with minimum-image velocity; orientations on SO(3) geodesics with body-frame angular-velocity target $u_R = \log(R_0^\top R_1)$, advanced by the exponential map. Conditional flow matching[4] with optimal-transport coupling[20]; RK sampling in 50 steps.

Deployable coset recovery. For space group with operations $\{(W_h, t_h)\}$ (pymatgen), each non-reference copy is assigned $\arg \min_h \|(c_m - (W_h c_0 + t_h)) \bmod 1\|$ under the minimum image, giving the generating operation index; deterministic (sg, h) ordering yields shared integer labels (238 total). This uses only centroids and space-group operations, never the observed rotation.

Training. Adam, lr 3×10^{-4} , gradient clipping, batch 16; default network $d=128$, four attention and four encoder layers; 800 steps for the coset runs, $K=4$ prior draws for the SO(3)-averaged objective. The predictor reuses the encoder and pair-bias trunk over conformer/centroid/lattice/space-group inputs (no orientation) and is trained to 39.5% held-out top-1.

B Per-seed numbers and tolerance sweep

Non-ref. loss drop, per seed. control 29.4/24.4/28.8%; deployable coset 43.2/37.2/42.9%; +SO(3)-avg 49.2/43.4/50.5%. **Orientation-isolated reconstruction** (coset-conditioned, best-of-eight), StructureMatcher (length, site) tolerance sweep: (0.2, 0.2) $\rightarrow$ 4.6%; (0.3, 0.5) $\rightarrow$ 27.5%; (0.5, 0.7) $\rightarrow$ 37.4%; oracle 100% throughout. **Predicted-coset** reconstruction: 0% match, 64.8° error (vs 41° template-supplied). **End-to-end** (all manifolds sampled, best-of-20): 0% match; median errors lattice-param 0.44, centroid 0.35, orientation 13.4°.